

RFP Use Case Extractor

Mentoring & Knowledge Transfer

How to understand, build and replicate the accelerator

1. What is the RFP Use Case Extractor?

Purpose

- AI-powered accelerator that automatically extracts use cases and requirements from large RFP(request for proposal) PDF documents.

Business Problem

- Manual analysis of RFPs is slow, effort intensive and error prone.
- Important requirements can be missed.
- Proposal teams spend significant time reading documents instead of solutioning.

Business Impact

- Faster bid response preparation
- Better requirement coverage
- Consistent documentation
- Complete traceability through page references

2. How the Solution Works

Input

- User uploads an RFP PDF.

Processing

- Document is analyzed using Gemini 2.5 Flash through LangChain.
- Prompt engineering guides the model to identify, classify and structure requirements.
- Every requirement is tagged with source page information.

Output

- Section → Sub-Section → Feature → Requirement hierarchy
- Excel report for analysis
- Word report for business consumption

4.1. AI-Enabled Crime Pattern Analytics			
Real-Time Crime Pattern Analytics			
	Real-Time Crime Pattern Analytics	Live crime feed dashboard reflecting current incidents and locations	Page 18
	Real-Time Crime Pattern Analytics	Hotspot detection using unsupervised ML algorithms like DBSCAN, K-means etc.	Page 18
	Real-Time Crime Pattern Analytics	Time-series pattern analysis for detecting trends e.g., seasonal spikes	Page 18

3. Architecture & Technology Stack

Architecture Flow

User → Streamlit UI → PDF Upload → LangChain → Gemini 2.5 Flash → Hierarchy Builder → Excel & Word Reports

Technology Stack

- Python
- Streamlit
- LangChain Google GenAI
- Gemini 2.5 Flash
- openpyxl for Excel generation
- python-docx for Word generation

Key Design Principle

Structured output with traceability back to source pages.

4. Secret Sauce - Prompt Engineering

Why It Matters

The model is not asked to summarize the RFP. It is instructed to extract requirements using a predefined hierarchy and strict rules.

Core Rules

- Extract explicit requirements only
- Maintain hierarchy relationships
- Include page references
- Keep business context intact
- Generate consistent output structure

Outcome

High quality, auditable and reusable requirement documentation.

5. Demo Flow & Replication Guide

Demo Flow

1. Upload PDF
2. Run extraction
3. Review hierarchy
4. Generate Excel report
5. Generate Word report
6. Validate source references

How Team Can Replicate

- Reuse architecture pattern
- Create robust prompts
- Modularize extraction and reporting layers
- Add improvements such as HITL, etc.

Key Learning

Transform unstructured RFP documents into structured business assets using multimodal AI.